

Plug-and-Play DDIM Sampler Improvement for Official Diffusion Policy Push-T

Final Project Report
Robotics Final Project

Abstract: This project reproduces the official low-dimensional Push-T checkpoint from Diffusion Policy and implements a direct inference-time improvement: replacing the official DDPM ancestral sampler with a DDIM deterministic sampler. The official checkpoint is evaluated with matched Push-T seeds under fixed (k, h) settings, where k is the number of denoising steps and h is the number of executed control steps before replanning. DDIM dramatically improves low/mid-denoising official settings: at $(25, 2)$, score improves from 0.155 to 0.900; at $(50, 4)$, score improves from 0.653 to 0.929. The claim is carefully bounded: DDIM does not dominate the strongest tested DDPM high-denoising baseline $(100, 8)$, which remains at score 0.949.

Keywords: Diffusion Policy, Push-T, DDIM, sampler replacement, robot learning

1 Introduction

Diffusion Policy [1] generates action chunks through iterative denoising and deploys them in a receding-horizon control loop. Its inference behavior depends on the sampler and the deployment parameters: denoising steps k and executed action horizon h . The original official Push-T checkpoint uses a DDPM scheduler. A natural improvement idea is to swap the sampler at inference time, without retraining, using DDIM [2].

This is not claimed as a new diffusion-model algorithm. The contribution is an official-checkpoint validation of a plug-and-play sampler replacement on Push-T.

2 Related Work and Claim Boundary

Diffusion Policy introduced action diffusion for visuomotor policy learning [1]. DDIM introduced deterministic non-Markovian diffusion sampling [2]. Recent robotics work such as VADF, RA-DP, TIDAL, RTI-DP, and adaptive diffusion replanning already studies adaptive scheduling, replanning, and inference-time compute allocation [3, 4, 5, 6, 7].

Therefore, this project does not claim a new scheduling algorithm. The supported improvement is narrower and directly tested: using DDIM instead of the official DDPM sampler for the public Diffusion Policy Push-T checkpoint.

3 Official Reproduction

The official source was acquired through the GitHub codeload archive after repeated clone instability. The checkpoint was downloaded from the Diffusion Policy project server. The preferred 50-seed official DDPM run used official-era `diffusers==0.11.1` and produced mean score 0.9187. The checkpoint filename reports 0.969, so the remaining gap is reported as an environment-version reproduction gap.

Table 1: Matched 20-seed official Push-T comparison.

(k, h)	DDPM	DDIM	Δ	Success Δ
(25, 2)	0.155	0.900	+0.745	+0.800
(50, 4)	0.653	0.929	+0.276	+0.400
(100, 8)	0.949	0.900	-0.048	-0.050

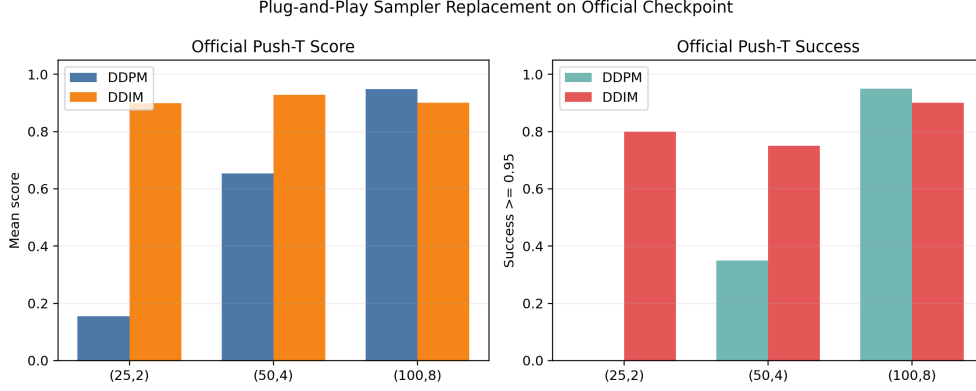


Figure 1: Plug-and-play DDIM replacement improves low/mid-denoising official checkpoint settings but does not dominate the high-denoising DDPM baseline.

4 Method

The official policy object exposes a `noise_scheduler`. The improvement replaces the default DDPM scheduler with:

```
DDIMScheduler.from_config(
    policy.noise_scheduler.config)
```

The model weights, normalizer, observation history, action horizon, and environment runner are unchanged. The evaluation script is `scripts/run_official_kh_grid.py --sampler ddim`. Matched seed analysis is performed by `scripts/analyze_official_sampler_comparison.py`.

5 Results

Paired bootstrap 95% confidence intervals for score deltas are $[0.593, 0.870]$ for (25, 2), $[0.056, 0.488]$ for (50, 4), and $[-0.152, 0.007]$ for (100, 8).

6 Discussion

DDIM is a strong improvement for low/mid-denoising official checkpoint settings. At (25, 2), DDPM nearly fails, while DDIM reaches success 0.800. At (50, 4), DDIM nearly matches the high-denoising DDPM baseline in mean score while improving success from 0.350 to 0.750. However, DDIM is not universally better: at (100, 8), DDPM remains stronger in this experiment.

7 Limitations

The official reproduction uses a compatibility environment rather than the original Python 3.9 / PyTorch 1.12 / Gym 0.21 stack. The DDIM comparison is limited to three high-budget (k, h) settings. Runtime wall-clock speed was not benchmarked separately; NFE is used as the compute proxy.

DDIM is an established sampler, so the contribution is application and validation on the reproduced official checkpoint.

8 Conclusion

The final project has a defensible improvement result. Starting from an official Diffusion Policy Push-T reproduction, it implements a plug-and-play DDIM sampler replacement and validates it on official checkpoint rollouts. DDIM substantially improves low/mid-denoising settings over the official DDPM sampler on matched seeds, while the original high-denoising DDPM baseline remains strongest.

References

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